

FAST Direct Hot-Pressing (FAST DHP), the industrial equivalent to SPS

SPS and Direct Hot-Pressing both belong to the generic group of Field Assisted Sintering Techniques (FAST).

When applying a standard (unpulsed) current, the technique is often referred to as “FAST Direct Hot-Pressing (FAST DHP)”. Another common term is “Rapid Hot-Pressing (RHP)” or “Electric Resistance Sintering”. When applying a pulsed current, it is referred to as “Spark Plasma Sintering (SPS)”. The term “Spark Plasma Sintering” is controversial since there is no evidence of a spark or plasma (Wikipedia)¹.

Whereas the names can be considered to be a marketing tool of the manufacturers, latest research suggests that there is no evident difference between sintering with pulsed or unpulsed current (FAST SPS or FAST DHP). The same improved sinter results (compared to traditional sintering) can be achieved by all FAST sintering techniques.²

The main difference is the user group. FAST Direct Hot-Pressing is a sintering standard in several industries since the 1950ies with the diamond tool industry being a pioneer. Until today, Dr. Fritsch alone has sold more than 1.000 machines worldwide. Sintering with a pulsed current, on the other hand, is a variation of the Direct Hot-Pressing technique which appeared on the market decades later. The biggest European manufacturer of SPS machines has sold 128 machines until 2012³. All Japanese SPS-manufacturers together sold approx. 650 machines until 2012⁴.

A reason for the high market share of FAST DHP machines is the fact, that the machines have consequently been optimized for industrial mass production. FAST DHP is a sintering technique, which allows upscaling from R&D to fully automated mass production without changing the production technology. Another reason is the fact, that FAST DHP machines are available for a fraction of the cost of a FAST SPS system due to the much larger production volumes. This makes the FAST sintering technique attractive for completely new industrial applications :

- Friction materials like brakes for high speed trains, motorcycles, wind power generators and clutches for trucks, vessels and tractors.

Wear resistance materials like:

- Hard metals (pure W and WC plus Co)
- Functionally Graded Materials (FGMs), for example weldable WC
- High-performance ceramic materials
- Diamond tools (metal-diamond-alloys)

Coating materials (Sputter Targets) and Heat Sinks.

¹ Wikipedia-Article about “Spark Plasma Sintering” (as of 18.03.2014)

² „International Powder Metallurgy Directory“ (January 4th, 2012): *2011 Hagen Symposium. A Review of Spark Plasma Sintering* by Prof. Bernd Kieback, Director of Fraunhofer IFAM branch lab Dresden and the Institute for Materials Science at the Technical University of Dresden (Germany). Summary has been published by Dr. Georg Schlieper.

³ FCT Company Presentation as of 05.12.2012

⁴ Prof. T. Goto, Tohoku University Japan, cited a presentation of Prof. Dr. Guillon as of 19.06.2013

Conclusion

Field Assisted Sintering Techniques are the answer to many issues industries around the world are dealing with. They allow to sinter materials with exactly pre-defined characteristics and combine this precision with a very fast and energy-efficient manufacturing process. The development of new materials can be accelerated significantly compared to traditional hot-pressing techniques. In fact, the sinter quality and achievable density come close or is comparable to Hot-Isostatic-Pressing (HIP) but at a fraction of the cost and without the undesired grain growth. Among the FAST techniques, FAST Direct Hot-Pressing is the most commonly used technique.

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